

JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

RELATIONS & FUNCTIONS **PART-03 (131 to 245)**

DIBY (DO IT BY YOURSELF)

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RANGE OF FUNCTION

DIBY-131

The minimum value of $2^{\sin x} + 2^{\cos x}$ is:

[JEE Mains-2020]

- (A) $2^{-1+\sqrt{2}}$ (B) $2^{1-\sqrt{2}}$ (C) $2^{1-\frac{1}{\sqrt{2}}}$ (D) $2^{-1+\frac{1}{\sqrt{2}}}$

[Ans.: (C)]

DIBY-132

Let the range of the function

$f(x) = 6 + 16\cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x, x \in \mathbb{R}$ be $[\alpha, \beta]$. Then the distance of the point (α, β) from the line $3x + 4y + 12 = 0$ is:

[JEE Mains-2025]

- (A) 8 (B) 11 (C) 9 (D) 10

[Ans.: (B)]

DIBY-133

Let $f: (1, 3) \rightarrow \mathbb{R}$ be a function defined by $f(x) = \frac{x[x]}{1+x^2}$, where $[x]$ denotes the greatest integer $\leq x$. Then the range of f is:

[JEE Mains-2020]

- (A) $\left(\frac{3}{4}, \frac{4}{5}\right)$ (B) $\left(\frac{2}{5}, \frac{1}{2}\right) \cup \left(\frac{3}{5}, \frac{4}{5}\right]$ (C) $\left(\frac{2}{5}, \frac{3}{5}\right] \cup \left(\frac{3}{4}, \frac{4}{5}\right)$ (D) $\left(\frac{2}{5}, \frac{4}{5}\right]$

[Ans.: (B)]

RANGE OF FUNCTION

DIBY-134

If the range of $f(\theta) = \frac{\sin^4 \theta + 3\cos^2 \theta}{\sin^4 \theta + \cos^2 \theta}$, $\theta \in \mathbb{R}$ is $[\alpha, \beta]$, then sum of the infinite G.P., whose first term is 64 and the common ratio is $\frac{\alpha}{\beta}$, is equal to ____.

[JEE Mains-2024]

[Ans. : 96]

DIBY-135

Let $f(\theta) = \sin^2 \theta + \cos^2 \theta + \tan^2 \theta + \sec^2 \theta + \operatorname{cosec}^2 \theta + \cot^2 \theta$, then the minimum value of $f(\theta)$ is:

[Ans. : 7]

DIBY-136 The range of the function $f(x) = \frac{x^2}{x^4 + 1}$ is

- (A) $(0, 1/2)$ (B) $[0, 1/2]$ (C) $[0, \infty)$ (D) $[0, 2]$

[Ans. : (B)]

DIBY-137

Range of $f(x) = \frac{\tan(\pi[x^2 - x])}{1 + \sin x(\cos x)}$ is (where $[.]$ denotes the greatest integer function):

- (A) $(-\infty, \infty) \sim [0, \tan 1]$ (B) $(-\infty, \infty) \sim [\tan 2, 0)$
 (C) $[\tan 2, \tan 1]$ (D) None of these

[Ans. : (D)]

RANGE OF FUNCTION

DIBY-138

Let $f(x) = (x+3)^2(x-2)^3$, $x \in [-4, 4]$. If M and m are maximum and minimum values of f , respectively $[-4, 4]$, then the value of $M - m$ is [JEE Mains-2024]

- (A) 600 (B) 392 (C) 608 (D) 108

[Ans.: (C)]

DIBY-139

Let the sum of the maximum and the minimum values of the function

$f(x) = \frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$, where $\gcd(m, n) = 1$. Then $m + n$ is equal to

[JEE Mains-2024]

- (A) 195 (B) 201 (C) 217 (D) 182

[Ans.: (B)]

DIBY-140

Find range of $f(x) = \frac{e^{2x}-e^x+1}{e^{2x}+e^x+1}$, $x \in R$.

[Ans.: $[1/3, 1]$]

CLASSIFICATION OF FUNCTIONS



DIBY-141

Classify following function defined as:

(i) $f(x) = x^2 + \ln x$

(ii) $f(x) = 9^{|x|} + 9^{2-x}$

[Ans. : (i) $f(x)$ is bijective,
(ii) $f(x)$ is many-one and into function]

DIBY-142

$f: [0, 3] \rightarrow A$ such that $f(x) = x^2 - 2x + 5$. Find A if $f(x)$ is onto?

[Ans. : $A = [4, 8]$]



Classify following function as injective or surjective:

(i) $f: [-1, 3] \rightarrow [-8, 72]$ be defined as $f(x) = 4x^3 - 12x$,

(ii) $f: R \rightarrow \{x \in R: -1 < x < 1\}$, $f(x) = \frac{x}{1+|x|}$

(iii) $f: [-1, 3] \rightarrow [-37, 27]$, $f(x) = 2x^3 - 6x^2 - 18x + 17$

(iv) $f: R \rightarrow R$, $f(x) = e^{x^2} + \cos x$

(v) $f: R \rightarrow R$, $f(x) = 2x - \sin x$,

Ans.:

(i) many-one and onto

(ii) one-one and onto

(iii) one-one and onto

(iv) many-one and into

(v) one-one and onto

CLASSIFICATION OF FUNCTIONS

DIBY-144

The function $f(x) = \frac{x^2+2x-15}{x^2-4x+9}$, $x \in R$

[JEE Mains-2024]

- (A) both one-one and onto. (B) onto but not one-one.
(C) neither one-one nor onto. (D) one-one but not onto.

[Ans.: (c)]

DIBY-145

The function $f: N - \{1\} \rightarrow N$; defined by $f(n)$ = the highest prime factor of n , is:

[JEE Mains-2024]

- (A) Both one-one and onto (B) One-one only
(C) Onto only (D) Neither one-one nor onto

[Ans.: (D)]

DIBY-146

Let $f: R \rightarrow R$ be a function such that $f(x) = \frac{x^2+2x+1}{x^2+1}$. Then

[JEE Mains-2023]

- (A) $f(x)$ is many-one in $(-\infty, -1)$ (B) $f(x)$ is many-one in $(1, \infty)$
(C) $f(x)$ is one-one in $[1, \infty)$ but not in $(-\infty, \infty)$ (D) $f(x)$ is one-one in $(-\infty, \infty)$

[Ans.: (c)]

CLASSIFICATION OF FUNCTIONS

DIBY-147

The function $f: [0, 3] \rightarrow [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is: [JEE-2012]

- (A) one-one and onto (B) onto but not one-one
(D) one-one but not onto (D) neither one-one nor onto

[Ans.: (B)]

DIBY-148

Let a function $f: N \rightarrow N$ be defined by $f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \dots \\ n-1, & n = 3, 7, 11, 15, \dots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \dots \end{cases}$ then, f is

[JEE Mains-2022]

- (A) One-one but not onto (B) Onto but not one-one
(C) Neither one-one nor onto (D) One-one and onto

[Ans.: (D)]

CLASSIFICATION OF FUNCTIONS

DIBY-149

Let N be the set of natural numbers and two functions f and g be defined as

$f, g: N \rightarrow N$ such that $f(n) = \begin{cases} \frac{n+1}{2} & \text{if } n \text{ is odd} \\ \frac{n}{2} & \text{if } n \text{ is even} \end{cases}$ and $g(n) = n - (-1)^n$. Then fog is:

[JEE Mains-2019]

- | | |
|----------------------------|------------------------------|
| (A) Onto but not one-one. | (B) One-one but not onto. |
| (C) Both one-one and onto. | (D) Neither one-one nor onto |

[Ans. : (A)]

DIBY-150

Let $A = \{x \in R: x \text{ is not a positive integer}\}$. Define a function $f: A \rightarrow R$ as $f(x) = \frac{2x}{x-1}$

then f is

[JEE Mains-2019]

- | | |
|----------------------------------|--------------------------------------|
| (A) Injective but nor subjective | (B) Not injective |
| (C) Subjective but not injective | (D) Neither injective nor subjective |

[Ans. : (A)]

CLASSIFICATION OF FUNCTIONS

DIBY-151

Let $f: R \rightarrow R$ be defined as $f(x) = 2x - 1$ and $g: R - \{1\} \rightarrow R - \{1\}$ be defined as

$g(x) = \frac{x - \frac{1}{2}}{x - 1}$. Then the composition function $f(g(x))$ is

[JEE Mains-2021]

- | | |
|------------------------------|--------------------------|
| (A) Both one-one and onto | (B) Onto but not one-one |
| (C) Neither one-one nor onto | (D) One-one but not onto |

[Ans.: (D)]

DIBY-152

Let $f: R \rightarrow R$ defined as $f(x) = \frac{2x^2 - 3x + 2}{3x^2 + x + 3}$. Then f is

[JEE Mains-2026]

- | | |
|---------------------------|------------------------------|
| (A) Both one-one and onto | (B) One-one but not onto |
| (C) Onto but not one-one | (D) neither one-one nor onto |

[Ans.: (D)]

CLASSIFICATION OF FUNCTIONS



DIBY-153

The function $f: (-\infty, \infty) \rightarrow (-\infty, 1)$, defined by $f(x) = \frac{2^x - 2^{-x}}{2^x + 2^{-x}}$ is **[JEE Mains-2025]**

(A) Onto but not one-one

(B) Both one-one and onto

(C) One-one but not onto

(D) Neither one-one nor onto

[Ans.: (c)]

NUMBER OF FUNCTION CASE MAKING / CRITICAL THINKING

DIBY-154

Let $A = \{1, 3, 7, 9, 11\}$ and $B = \{2, 4, 5, 7, 8, 10, 12\}$. Then the total number of one-one maps $f: A \rightarrow B$, such that $f(1) + f(3) = 14$ is: [JEE Mains-2024]

- (A) 180 (B) 120 (C) 480 (D) 240

[Ans. : (D)]

DIBY-155

Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{1, 2, 3, 4, 5, 6\}$. Then the number of functions $f: A \rightarrow B$ satisfying $f(1) + f(2) = f(4) - 1$ is equal to [JEE Mains-2023]

[Ans. : 360]

DIBY-156

The number of bijective functions $f: \{1, 3, 5, 7, \dots, 99\} \rightarrow \{2, 4, 6, 8, \dots, 100\}$, such that $f(3) \geq f(9) \geq f(15) \geq f(21) \geq \dots \geq f(99) \dots$ [JEE Mains-2022]

- (A) ${}^{50}P_{17}$ (B) ${}^{50}P_{33}$ (C) $33! \times 17!$ (D) $\frac{50!}{2}$

[Ans. : (c)]

NUMBER OF FUNCTION CASE MAKING / CRITICAL THINKING

DIBY-157

Let $S = \{1, 2, 3, 4\}$. Then the number of elements in the set $\{f: S \times S \rightarrow S: f \text{ is onto and } f(a, b) = f(b, a) \geq a \forall (a, b) \in S \times S\}$ is **[JEE Mains-2022]**
[Ans. : 37]

DIBY-158

The total number of functions, $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6\}$ such that $f(1) + f(2) = f(3)$, is equal to: **[JEE Mains-2022]**
(A) 60 (B) 90 (C) 108 (D) 126
[Ans. : (B)]

DIBY-159

Let $A = \{1, 2, 3, 5, 8, 9\}$. Then the number of possible functions $f: A \rightarrow A$ such that $f(m \cdot n) = f(m) \cdot f(n)$ for every $m, n \in A$ with $m \cdot n \in A$ is equal to **[JEE Mains-2023]**
[Ans. : 432]

NUMBER OF FUNCTION CASE MAKING / CRITICAL THINKING

DIBY-160

Let $R = \{a, b, c, d, e\}$ and $S = \{1, 2, 3, 4\}$. Total number of onto function $f: R \rightarrow S$ such that $f(a) \neq 1$, is equal to
[JEE Mains-2023]
[Ans.: 180]

DIBY-161

Let $A = \{1, 2, 3, 4\}$ and $B = \{1, 4, 9, 16\}$. Then the number of many-one functions $f: A \rightarrow B$ such that $1 \in f(A)$ is equal to
[JEE Mains-2025]
 (A) 163 (B) 139 (C) 151 (D) 127
[Ans.: (c)]

DIBY-162

Consider two sets $A = \{x \in \mathbb{Z}: (|x - 3| - 3) \leq 1\}$ and
 $B = \left\{x \in \mathbb{R} - (1, 2): \frac{(x-2)(x-4)}{x-1} \log_e(|x - 2|) = 0\right\}$. Then the number of onto function $f: A \rightarrow B$ is equal to
[JEE Mains-2026]
 (A) 62 (B) 32 (C) 81 (D) 79
[Ans.: (A)]



NUMBER OF FUNCTION
CASE MAKING / CRITICAL THINKING

DIBY-163

Let $A = \{(x, y): 2x + 3y = 23, x, y \in N\}$ and $B = \{x: (x, y) \in A\}$. Then the number of one-one functions from A to B is equal to _____

[JEE Mains-2024]

[Ans. : 24]

DIBY-164

The number of functions f , from the set $A = \{x \in N: x^2 - 10x + 9 \leq 0\}$ to the set $B = \{n^2: n \in N\}$ such that $f(x) \leq (x - 3)^2 + 1$, for every $x \in A$, is ____

[JEE Mains-2022]

[Ans. : 1440]

DIBY-165

The number of functions $f: \{1, 2, 3, 4\} \rightarrow \{a, b, c\}$, which are not onto, is:

[JEE Mains-2026]

(A) 48

(B) 45

(C) 51

(D) 35

[Ans. : (B)]



**NUMBER OF FUNCTION
CASE MAKING / CRITICAL THINKING**

DIBY-166

Let $A = \{0, 1, 2, 3, 4, 5, 6, 7\}$. Then the number of bijective functions $f: A \rightarrow A$ such that $f(1) + f(2) = 3 - f(3)$ is equal to ____

[JEE Mains-2021]

[Ans. : 720]

COMPOSITE FUNCTION



DIBY-167

If $g(x) = 3x^2 + 2x - 3$, $f(0) = -3$ and $4g(f(x)) = 3x^2 - 32x + 72$, then $f(g(2))$ is equal to: **[JEE Mains-2026]**

- (A) $\frac{25}{6}$ (B) $-\frac{7}{2}$ (C) $\frac{7}{2}$ (D) $-\frac{25}{6}$

[Ans.: (C)]

DIBY-168

Let $f(x) = \frac{x-1}{x+1}$, $x \in \mathbb{R} - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$ for all $n \in \mathbb{N}$, then $f^6(6) + f^7(7)$ is equal to **[JEE Mains-2022]**

- (A) $\frac{7}{6}$ (B) $-\frac{3}{2}$ (C) $\frac{7}{12}$ (D) $-\frac{11}{12}$

[Ans.: (B)]

DIBY-169

If $f(x) = \frac{4x+3}{6x-4}$, $x \neq \frac{2}{3}$ and $(f \circ f)(x) = g(x)$, where $f: \mathbb{R} - \left\{\frac{2}{3}\right\} \rightarrow \mathbb{R} - \left\{\frac{2}{3}\right\}$, then $(g \circ g \circ g)(4)$ is equal to **[JEE Mains-2024]**

- (A) $-\frac{19}{20}$ (B) $\frac{19}{20}$ (C) -4 (D) 4

[Ans.: (D)]

COMPOSITE FUNCTION



DIBY-170

If $g(x) = x^2 + x - 1$ and $(g \circ f)(x) = 4x^2 - 10x + 5$, then $f\left(\frac{5}{4}\right)$ is equal to

[JEE Mains-2020]

- (A) $\frac{1}{2}$ (B) $\frac{-3}{2}$ (C) $\frac{-1}{2}$ (D) $\frac{3}{2}$

[Ans.: (C)]

DIBY-171

Let $f(x) = x^2 + 9$, $g(x) = \frac{x}{x-9}$ and $a = f \circ g(10)$, $b = g \circ f(3)$. If e and l denote the

eccentricity and the length of the latus rectum of the ellipse $\frac{x^2}{a} + \frac{y^2}{b} = 1$, then

$8e^2 + 1^2$ is equal to.

[JEE Mains-2024]

- (A) 16 (B) 8 (C) 6 (D) 12

[Ans.: (B)]

DIBY-172

Let $f: R - \left\{\frac{\alpha}{6}\right\} \rightarrow R$ be defined by $f(x) = \frac{5x+3}{6x-\alpha}$. Then the value of α for which $(f \circ f)(x) =$

x , for all $x \in R - \left\{\frac{\alpha}{6}\right\}$, is :

[JEE Mains-2021]

- (A) 6 (B) 8 (C) No such α exists (D) 5

[Ans.: (D)]

COMPOSITE FUNCTION



DIBY-173

For a suitably chosen real constant a , let a function, $f: R - \{-a\} \rightarrow R$ be defined by $f(x) = \frac{a-x}{a+x}$. Further suppose that for any real number $x \neq -a$ and

$f(x) \neq -a$, $(f \circ f)(x) = x$. Then $f\left(-\frac{1}{2}\right)$ is equal to

[JEE Mains-2020]

(A) -3

(B) $\frac{1}{3}$

(C) $-\frac{1}{3}$

(D) 3

[Ans.: (D)]

DIBY-174

For $x \in R - [0, 1]$, let $f_1(x) = \frac{1}{x}$, $f_2(x) = 1 - x$ and $f_3(x) = \frac{1}{1-x}$ be three given functions. If a function, $J(x)$ satisfies $(f_2 \circ J \circ f_1)(x) = f_3(x)$, then $J(x)$ is equal to

[JEE Mains-2019]

(A) $f_3(x)$

(B) $\frac{1}{x}f_3(x)$

(C) $f_2(x)$

(D) $f_1(x)$

[Ans.: (A)]

COMPOSITE FUNCTION



DIBY-175

Let $f(x) = x^2, x \in R$. For any $A \subseteq R$, define $g(A) = \{x \in R: f(x) \in A\}$. If $S = [0, 4]$, then which one of the following statements is not true? **[JEE Mains-2019]**

- (A) $f(g(S)) \neq f(S)$ (B) $f(g(S)) = S$
(C) $g(f(S)) = g(S)$ (D) $g(f(S)) \neq S$

[Ans.: (C)]

DIBY-176

Let $f(x) = e^x - x$ and $g(x) = x^2 - x, \forall x \in R$. Then the set of all $x \in R$, where the function $h(x) = (f \circ g)(x)$ is increasing, is: **[JEE Mains-2019]**

- (A) $\left[-1, \frac{-1}{2}\right] \cup \left[\frac{1}{2}, \infty\right)$ (B) $\left[0, \frac{1}{2}\right] \cup [1, \infty)$
(C) $\left[\frac{-1}{2}, 0\right] \cup [1, \infty)$ (D) $[0, \infty)$

[Ans.: (B)]

COMPOSITE FUNCTION



DIBY-177

For $x \in \left(0, \frac{3}{2}\right)$, let $f(x) = \sqrt{x}$, $g(x) = \tan x$ and $h(x) = \frac{1-x^2}{1+x^2}$. If $\phi(x) = (h \circ f \circ g)(x)$, then

$\phi\left(\frac{\pi}{3}\right)$ is equal to:

[JEE Mains-2019]

(A) $\tan \frac{11\pi}{12}$

(B) $\tan \frac{\pi}{12}$

(C) $\tan \frac{5\pi}{12}$

(D) $\tan \frac{7\pi}{12}$

[Ans.: (A)]

DIBY-178

Let $f, g: (1, \infty) \rightarrow \mathbb{R}$ be defined as $f(x) = \frac{2x+3}{5x+2}$ and $g(x) = \frac{2-3x}{1-x}$. If the range of the function $f \circ g: [2, 4] \rightarrow \mathbb{R}$ is $[\alpha, \beta]$, then $\frac{1}{\beta-\alpha}$ is equal to

(A) 68

(B) 29

(C) 2

(D) 56

[Ans.: (D)]

COMPOSITE FUNCTION



DIBY-179

Consider the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{2x}{\sqrt{1+9x^2}}$ If the composition of

$f(f \circ f \circ f \circ \dots \circ f)(x) = \frac{2^{10}x}{\sqrt{1+9\alpha x^2}}$, then the value of $\sqrt{3\alpha + 1}$ is equal to
.....10 Times.....

[JEE Mains-2024]

[Ans. : 1024]

DIBY-180

Let α, β and γ be three positive real numbers. Let $f(x) = \alpha x^5 + \beta x^3 + \gamma x, x \in R$ and $g: R \rightarrow R$ be such that $g(f(x)) = x$ for all $x \in R$. If $a_1, a_2, a_3, \dots, a_n$ be in arithmetic progression with mean zero, then the value of $f\left(g\left(\frac{1}{n} \sum_{i=1}^n f(a_i)\right)\right)$ is equal to:

[JEE Mains-2022]

(A) 0

(B) 3

(C) 9

(D) 27

[Ans. : (A)]

COMPOSITE FUNCTION



DIBY-181

Let $f: R - \left\{-\frac{1}{2}\right\} \rightarrow R$ and $g: R - \left\{-\frac{5}{2}\right\} \rightarrow R$ be defined as $f(x) = \frac{2x+3}{2x+1}$ and $g(x) = \frac{|x|+1}{2x+5}$.

Then the domain of the function $f \circ g$ is

[JEE Mains-2024]

- (A) $R - \left\{-\frac{5}{2}\right\}$ (B) R (C) $R - \left\{-\frac{7}{4}\right\}$ (D) $R - \left\{-\frac{5}{2}, -\frac{7}{4}\right\}$

[Ans.: (A)]

DIBY-182

Let $f(x) = \log_e x$ and $g(x) = \frac{x^4 - 2x^3 + 3x^2 - 2x + 2}{2x^2 - 2x + 1}$. Then, the domain of $f \circ g$ is

[JEE Mains-2025]

- (A) $(0, \infty)$ (B) $[0, \infty)$ (C) $[1, \infty)$ (D) R

[Ans.: (D)]

COMPOSITE FUNCTION



DIBY-183

Let $f: R \rightarrow R$ be defined as $f(x) = x - 1$ and $g: R - \{1, -1\} \rightarrow R$ be defined as

$g(x) = \frac{x^2}{x^2 - 1}$. Then the function $f \circ g$ is

[JEE Mains-2022]

- | | |
|------------------------------------|---------------------------------------|
| (A) One-one but not onto function | (B) Onto but not one-one function |
| (C) Both one-one and onto function | (D) Neither one-one nor onto function |

[Ans.: (D)]

DIBY-184

Let $f(x) = \begin{cases} x+1; & x \leq 1 \\ 2x+1; & 1 < x \leq 2 \end{cases}$ and $g(x) = \begin{cases} x^2; & -1 \leq x < 2 \\ x+2; & 2 \leq x \leq 3 \end{cases}$. Find $f \circ g$ and $g \circ f$.

$$[\text{Ans. : } f(g(x)) = \begin{cases} x^2 + 1; & -1 \leq x \leq 1 \\ 2x^2 + 1; & 1 < x \leq \sqrt{2} \end{cases}$$

$$g \circ f(x) = (x+1)^2, -2 \leq x < 1]$$

DIBY-185

Let $f: R \rightarrow R$ and $g: R \rightarrow R$ be defined as $f(x) = \begin{cases} \log_e x, & x > 0 \\ e^{-x}, & x \leq 0 \end{cases}$ and

$g(x) = \begin{cases} x, & x \geq 0 \\ e^x, & x < 0 \end{cases}$. Then $g \circ f: R \rightarrow R$ is

[JEE Mains-2024]

- (A) one-one but not onto
(C) onto but not one-one

- (B) neither one-one nor onto
(D) both one-one and onto

[Ans. : (B)]

DIBY-186

Let $A = \{1, 2, 3, \dots, 10\}$ and $f: A \rightarrow A$ be defined as $f(k) = \begin{cases} k+1, & \text{if } k \text{ is odd} \\ k, & \text{if } k \text{ is even} \end{cases}$. Then the number of possible functions $g: A \rightarrow A$ such that $g \circ f = f$ is **[JEE Mains-2021]**

(A) 10^5 (B) ${}^{10}C_5$ (C) 5^5 (D) $5!$

[Ans.: (A)]

DIBY-187

Let f and g be functions given by

$$f(x) = -1 + |x - 1|, \quad -1 \leq x \leq 3 \quad \text{and} \quad g(x) = 2 - |x + 1|, \quad -2 \leq x \leq 2$$

Find an explicit representation of $f \circ g$ and $g \circ f$.

$$\begin{aligned} \text{[Ans.: } (f \circ g)x &= \begin{cases} x+1, & -2 \leq x < -1 \\ -x-1, & -1 \leq x \leq 0 \\ x-1, & 0 < x \leq 2 \end{cases} \\ (g \circ f)x &= \begin{cases} 1+x, & -1 \leq x < 1 \\ 3-x, & 1 \leq x \leq 3 \end{cases} \end{aligned}$$

COMPOSITE FUNCTION



DIBY-188

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions and $g \circ f : A \rightarrow C$.

Which of the following statements is true?

- (A) If $g \circ f$ is one-one, then f and g both are one-one.
- (B) If $g \circ f$ is one-one, then f is one-one.
- (C) If $g \circ f$ is a bijection, then f is one-one and g is onto.
- (D) If f and g are both one-one, then $g \circ f$ is one-one.

[Ans. : B, C, D]

DIBY-189

Let $f_1(x) = x$, $f_2(x) = 1 - x$; $f_3(x) = \frac{1}{x}$, $f_4(x) = \frac{1}{1-x}$; $f_5(x) = \frac{x}{x-1}$; $f_6(x) = \frac{x-1}{x}$. Suppose that $f_6(f_m(x)) = f_4(x)$ and $f_n(f_4(x)) = f_3(x)$, then find the value of m and n .

[Ans. : $m = 6, n = 5$]

INVERSE OF FUNCTION



DIBY-190

Let $f: \mathbb{R} - \{3\} \rightarrow \mathbb{R} - \{1\}$ be defined by $f(x) = \frac{x-2}{x-3}$. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given as $g(x) = 2x - 3$. Then the sum of all the values of x for which $f^{-1}(x) + g^{-1}(x) = \frac{13}{2}$ is equal to

[JEE Mains-2021]

(A) 7

(B) 2

(C) 5

(D) 3

[Ans. : (C)]

DIBY-191

For some $a, b, c \in \mathbb{N}$, let $f(x) = ax - 3$ and $g(x) = x^b + c$, $x \in \mathbb{R}$. If $(f \circ g)^{-1}(x) = \left(\frac{x-7}{2}\right)^{1/3}$, then $(f \circ g)(ac) + (g \circ f)(b)$ is equal to

[JEE Mains-2023]

[Ans. : 2039]

DIBY-192

Inverse of $f(x) = \begin{cases} x, & x < 1 \\ x^2, & 1 \leq x \leq 4 \\ 8\sqrt{x}, & x > 4 \end{cases}$ is

$$[\text{Ans.} : f^{-1}(x) = \begin{cases} x, & x < 1 \\ \sqrt{x}, & 1 \leq x \leq 16 \\ \frac{x^2}{64}, & x > 16 \end{cases}]$$

DIBY-193

Prove that $f(x) = e^x - e^{-x}$ is invertible and also find inverse of $f(x)$.

$$[\text{Ans.} : f^{-1}(x) = \ln \left(\frac{x + \sqrt{x^2 + 4}}{2} \right)]$$

DIBY-194

If $f(x) = \frac{x|x|}{1+x^2}$ then show that $f^{-1}(x) = \text{sgn}(x) \cdot \sqrt{\frac{|x|}{1-|x|}}$, $-1 < x < 1$.

INVERSE OF FUNCTION



DIBY-195

If $f: [1, \infty) \rightarrow [2, \infty)$ is given by $x + \frac{1}{x}$, then $f^{-1}(x)$ equals

[IIT-JEE 2001]

(A) $\frac{x + \sqrt{x^2 - 4}}{2}$

(B) $\frac{x}{1 + x^2}$

(C) $\frac{x - \sqrt{x^2 - 4}}{2}$

(D) $1 + \sqrt{x^2 - 4}$

[Ans.: (A)]

DIBY-196

Suppose $f(x) = (x + 1)^2$ for $x \geq -1$. If $g(x)$ is the function whose graph is reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals [IIT-JEE 2002]

(A) $-\sqrt{x} - 1, x \geq 0$ (B) $\frac{1}{(x+1)^2}, x > -1$ (C) $\sqrt{x+1}, x \geq -1$ (D) $\sqrt{x} - 1, x \geq 0$

[Ans.: (D)]

DIBY-197

Let f be a one-one function with domain $\{x, y, z\}$ and range $\{1, 2, 3\}$. It is given that exactly one of the following statements is true and the remaining two are false

$f(x) = 1, f(y) \neq 1, f(z) \neq 2$ determine $f^{-1}(1)$.

[IIT-JEE 1982]

[Ans.: y]

INVERSE OF FUNCTION



DIBY-198

Let $f(x) = \frac{ax}{x+1}$, $x \neq -1$. Then, for what value of a is $f(f(x)) = x$? [IIT-JEE 2001]

- (A) $-\sqrt{2}$ (B) $\sqrt{2}$ (C) 1 (D) -1

[Ans.: (D)]

DIBY-199

If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g\{f(x)\}$ is invertible in the domain [IIT-JEE 2004]

- (A) $\left[0, \frac{\pi}{2}\right]$ (B) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (C) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (D) $[0, \pi]$

[Ans.: (B)]

DIBY-200

Let function $f: R \rightarrow R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then, f is [IIT-JEE 2002]

- (A) one-to-one and onto (B) one-to-one but not onto
(C) onto but not one-to-one (D) neither one-to-one nor onto

[Ans.: (A)]

DIBY-201

There are exactly two distinct linear functions _____ and _____ which map $\{-1, 1\}$ on to $\{0, 2\}$. **[IIT-JEE 1989]**

[Ans. : $f(x) = \pm x + 1$]

DIBY-202

Consider the following functions in their default domains:

- (i) $\frac{2^x-1}{2^x+1}$ (ii) $\frac{x^2+1}{x \sin x}$ (iii) $\ln \left(\frac{1+x}{1-x} \right)$ (iv) $xe^{|x|+\cos x}$

Which of these is/are odd:

- (A) (i) & (iii) (B) (i) & (iv) (C) All four (D) (i), (iii) & (iv)

[Ans. (D)]

DIBY-203

Let $f: [-4, 4] \sim \{-\pi, 0, \pi\} \rightarrow \mathbf{R}$ such that $f(x) = \cot(\sin x) + \left[\frac{x^2}{|a|} \right]$, where $[.]$ denotes the greatest integer function is an odd function, complete set of values of ' a ' is:

- (A) $(-16, 16) \sim \{0\}$ (B) $(-\infty, -16) \cup (16, \infty)$
 (C) $[-16, 16] \sim \{0\}$ (D) $(-\infty, -16) \cup [16, \infty)$

[Ans. (B)]

EVEN & ODD FUNCTIONS



DIBY-204

Let $f(x)$ be an odd function defined on $\mathbb{R}$ such that $f(1) = 2$, $f(3) = 5$ and

$f(-5) = -1$. The value of $\frac{f(f(f(-3))) + f(f(0))}{3f(1) - 2f(3) - f(5)}$ is:

(A) $-\frac{2}{5}$

(B) $-\frac{2}{3}$

(C) $\frac{2}{5}$

(D) 1

[Ans. (C)]

DIBY-205

f is an even function defined on the interval $(-5, 5)$, then four real values of x satisfying the equation $f(x) = f\left(\frac{x+1}{x+2}\right)$ are

[IIT-JEE 1996]

[Ans.: $x = \frac{\pm 3 \pm \sqrt{5}}{2}$]

DIBY-206

Let $f(x)$ be an odd function defined on $(-\infty, \infty)$ and $g(x)$ be an even function defined on $(-\infty, \infty)$. If $f(x) - g(x) = x^2 + 5x + 7$, then $g(2)$ is equal to

[Ans. -11]

DIBY-207

Let $f(x) = e^{\{e^{|x|}\operatorname{sgn} x\}}$ and $g(x) = e^{[e^{|x|}\operatorname{sgn} x]}$, $x \in \mathbb{R}$ where $\{x\}$ and $[]$ denotes the fractional part and integral part functions respectively.

Also, $h(x) = \ln(f(x)) + \ln(g(x))$, then for all real x , $h(x)$ is:

- (A) an odd function
- (B) an even function
- (C) neither an odd nor an even function
- (D) both odd as well as even function

[Ans. (A)]

DIBY-208

If $h(x) = A x^5 + B \sin x + C \ln \left(\frac{1+x}{1-x} \right) + 7$, where A, B, C are non-zero real constants and $h\left(\frac{-1}{2}\right) = 6$, then find the value of $h\left(\frac{\operatorname{sgn}(e^{-x})}{2}\right)$.

[Ans. 8]

DIBY-209

Let $f(x) = a^x (a > 0)$ be written as $f(x) = f_1(x) + f_2(x)$, where $f_1(x)$ is an even function and $f_2(x)$ is an odd function. Then $f_1(x+y) + f_1(x-y)$ equals is:

- | | |
|-------------------------|---------------------|
| (A) $2f_1(x+y)f_2(x-y)$ | (B) $2f_1(x)f_1(y)$ |
| (C) $2f_1(x+y)f_1(x-y)$ | (D) $2f_1(x)f_2(y)$ |

[Ans. (B)]

DIBY-210

Four statements, about the graph of $f(x) = \log_{10} \frac{1+x}{1-x}$, are given below.

Mark all the correct statements:

- I: the graph is symmetric about the y -axis.
- II: the graph lies in the 4th and 1st quadrants.
- III: the graph is symmetric about the origin.
- IV: the graph lies in the 3rd and 1st quadrants and goes through the origin.

- | | | | |
|--------------|----------------|---------|----------------|
| (A) I and IV | (B) II and III | (C) III | (D) III and IV |
|--------------|----------------|---------|----------------|

[Ans. (D)]

MISCELLANEOUS QUESTIONS



DIBY-211

If the function $f: (-\infty, -1] \rightarrow (a, b]$ defined by $f(x) = e^{x^3-3x+1}$ is one-one and onto, then the distance of the point $P(2b + 4, a + 2)$ from the line $x + e^{-3}y = 4$ is:

[JEE Mains-2024]

- (A) $2\sqrt{1 + e^6}$ (B) $4\sqrt{1 + e^6}$ (C) $3\sqrt{1 + e^6}$ (D) $\sqrt{1 + e^6}$

[Ans. (A)]

DIBY-212

Let $A = \{a_1, a_2, a_3, a_4\}$ & $B = \{b_1, b_2, b_3, b_4, b_5\}$. The number of injective functions f from A to B such that $f(a_1) \neq b_4$ and $f(a_2) = b_1$, is

[Ans. 18]

DIBY-213

Number of functions $f: \{1, 2, \dots, 100\} \rightarrow \{0, 1\}$, that assign 1 to exactly one of the positive integers less than or equal to 98, is equal to _____.

[JEE Mains-2025]

[Ans. 392]

MISCELLANEOUS QUESTIONS

DIBY-214

Consider the sets $A = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 = 25\}$,

$B = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + 9y^2 = 144\}$,

$C = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x^2 + y^2 \leq 4\}$,

and $D = A \cap B$. The total number of one-one functions from the set D to the set C is:

(A) 15120

(B) 19320

(C) 17160

(D) 18290

[Ans. (c)]

DIBY-215

Let $S = \{1, 2, 3, 4, 5, 6\}$. Then the number of one-one functions $f: S \rightarrow P(S)$, where $P(S)$ denote the power set of S , such that $f(n) \subset f(m)$ where $n < m$ is _____.

[JEE Mains-2023]

[Ans. 3240]

DIBY-216

Let a function f be defined as $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$. If f satisfy $f(f(x)) = f(x)$, $\forall x \in \{1, 2, 3, 4\}$, the number of such functions is _____.

[Ans. 41]

MISCELLANEOUS QUESTIONS



DIBY-217

Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6, 7\}$ are two sets. Let m is the number of one-one functions $f: A \rightarrow B$ such that $f(i) \neq i \forall i \in A$ and n is the number of one-one functions $f: A \rightarrow B$ such that $|f(i) - i| \leq 3 \forall i \in A$, then find the value of $(m + n)$.

[Ans. 94]

DIBY-218

If $f(x) = \sin^2 x + \sin^2 \left(x + \frac{\pi}{3}\right) + \cos x \cos \left(x + \frac{\pi}{3}\right)$ and $g\left(\frac{5}{4}\right) = 1$, then find $(g \circ f)(x)$:

[IIT-JEE 1996]

[Ans. 1]

DIBY-219

Let X be a set with exactly 5 elements and Y be a set with exactly 7 elements. If α is the number of one-one functions from X to Y and β is the number of onto functions from Y to X , then the value of $\frac{1}{5!}(\beta - \alpha)$ is ____.

[JEE (Adv.)-2018, Paper-2]

[Ans. 119]

DIBY-220

Column-I

(A) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$f(x) = e^{\operatorname{sgn} x} + e^{x^2}, \text{ where } \operatorname{sgn} x$$

denotes signum function of x , then $f(x)$ is

(B) Let $f : (-1, 1) \rightarrow \mathbb{R}$ defined as $f(x) = x[x^4] + \frac{1}{\sqrt{1-x^2}}$

where $[x]$ denotes greatest integer less than or equal to x , then $f(x)$ is:

(C) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as $f(x) = \frac{x(x+1)(x^4+1)+2x^4+x^2+2}{x^2+x+1}$,

then $f(x)$ is:

(D) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$f(x) = x + 3x^3 + 5x^5 + \dots + 101x^{101}, \text{ then } f(x) \text{ is:}$$

Column-II

(P) Odd

(Q) Even

(R) Neither odd nor even

(S) One-One

(T) Many-One

[Ans. (A) $\rightarrow$ R, T; (B) $\rightarrow$ Q, T; (C) $\rightarrow$ R, T; (D) $\rightarrow$ P, S]

COMPOSITE OF NON UNIFORMLY DEFINED FUNCTION



DIBY-221 (Paragraph for Question Nos. 221(i) to 221(iii))

Let $f(x) = \begin{cases} \beta x^2 + 3, & -\infty < x < -1 \\ 2x + \alpha, & -1 \leq x < \infty \end{cases}$ and $g(x) = \begin{cases} x + 4, & 0 \leq x \leq 8 \\ -3x - 2, & -\infty < x < 0 \end{cases}$.

(i) The function $g(f(x))$ is not defined if :

(A) $\alpha \in (10, \infty), \beta \in (5, \infty)$

(B) $\alpha \in (4, 10), \beta \in (5, \infty)$

(C) $\alpha \in (10, \infty), \beta \in (0, 1)$

(D) $\alpha \in (4, 10), \beta \in (1, 5)$

(ii) If $\alpha = 2$ and $\beta = 3$, then range of $g(f(x))$ is equal to :

(A) $(-2, 12]$

(B) $(0, 12]$

(C) $[4, 12]$

(D) $[-1, 12]$

(iii) If $\alpha = 3$, then the value of x in interval $[1, 3]$ for which $f(x) + g(x) = 12$, is

(A) 1

(B) $\frac{3}{2}$

(C) $\frac{5}{3}$

(D) 3

[Ans. : (i) (A), (ii) (C), (iii) (C)]

DIBY-222

Find the period of the following function:

(i) $f(x) = \operatorname{cosec}^2 3x + \cot 4x$

(ii) $f(x) = \frac{1}{2} \left(\frac{|\sin x|}{\cos x} + \frac{\sin x}{|\cos x|} \right)$

(iii) $f(x) = \tan \frac{\pi}{2} [x]$, where $[.]$ denotes greatest integer ser function

(iv) $f(x) = |\cos \pi x| + |\cos 2\pi x| + \cdots + |\cos n\pi x| + e^{x-[x]}$ where $[.]$ is greatest integer function.

(v) $f(x) = \operatorname{sgn}([x] + [-x])$ where $[.]$ denotes greatest integer function

(vi) $f(x) = \exp(x - [x] + \sqrt{x - [x]} + (x - [x])^2 + |\sin \pi x| + |\cos \pi x| + |\tan \pi x|)$ where $[.]$ is GIF

(vii) $f(x) = \sin 3x \cos [3x] - \cos 3x \sin [3x]$, where $[.]$ denotes the greatest integer function

[Ans. : (i) π ; (ii) 2π ; (iii) 2; (iv) 1; (v) 1; (vi) 1; (vii) $1/3$]

PERIODIC FUNCTION



DIBY-223

A real valued function f satisfies $f(10 + x) = f(10 - x)$ and $f(20 - x) = -f(20 + x)$, for all $x \in R$ show that f is an odd function as well as periodic function.

DIBY-224

Find the value of $n \in I$ for which the function $f(x) = \frac{\sin nx}{\sin(x/n)}$ has 4π as its period.

FUNCTIONAL EQUATIONS

[Ans. : 2]

DIBY-225

If $3f(x) - f\left(\frac{1}{x}\right) = \log_e x^4$ for $x > 0$, then find $f(e^x)$.

[Ans. : x]

DIBY-226

If $x = \log_4 \left(\frac{2f(x)}{1-f(x)} \right)$, then find $f(2010) + f(-2009)$.

[Ans. : 1]

FUNCTIONAL EQUATIONS



DIBY-227

Let $f(x)$ be an even function satisfying $f(x) + f(x+2) = f(x+1) \forall x \in R$, where $f(-13) = 4$, then find $f(-5) + f(1)$.

[Ans. : 8]

DIBY-228

Suppose f is a function satisfying $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{N}$ and $f(1) = \frac{1}{5}$.

If $\sum_{n=1}^m \frac{f(n)}{n(n+1)(n+2)} = \frac{1}{12}$, then m is equal to

[JEE Mains-2023]

[Ans. : 10]

DIBY-229

A real valued function $f(x)$ satisfies the function equation

$f(x-y) = f(x)f(y) - f(a-x)f(a+y)$ where a is a given constant and $f(0) = 1$,

$f(2a-x)$ is equal to

[AIEEE-2005]

(A) $f(1) + f(a-x)$ (B) $f(-x)$ (C) $-f(x)$ (D) $f(x)$

[Ans. : (c)]

FUNCTIONAL EQUATIONS



DIBY-230

Let $f(x)$ be an even function satisfying $f(x) + f(x+2) = f(x+1) \forall x \in R$, where $f(-13) = 4$, then find $f(-5) + f(1)$.

[Ans.: 8]

DIBY-231

If a polynomial function ' f ' satisfies the relation

$$\log_2(f(x)) = \log_2\left(2 + \frac{2}{3} + \frac{2}{9} + \dots \infty\right) \cdot \log_3\left(1 + \frac{f(x)}{f\left(\frac{1}{x}\right)}\right) \text{ and } f(10) = 1001, \text{ then the}$$

value of $f(20)$ is

- (A) 2002 (B) 7999 (C) 8001 (D) 16001

[Ans.: (C)]

DIBY-232

Let $f: N \rightarrow N$ be a function such that $f(m+n) = f(m) + f(n)$ for every $m, n \in N$. If

$f(6) = 18$, then $f(2) \cdot f(3)$ is equal to

[JEE Mains-2021]

- (A) 6 (B) 54 (C) 18 (D) 36

[Ans.: (B)]

FUNCTIONAL EQUATIONS



DIBY-233

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in \mathbb{R}$, and $g: \mathbb{R} \rightarrow (0, \infty)$ be a function such that $g(x+y) = g(x)g(y)$ for all $x, y \in \mathbb{R}$. If $f\left(\frac{-3}{5}\right) = 12$ and $g\left(\frac{-1}{3}\right) = 2$, then the value of $\left(f\left(\frac{1}{4}\right) + g(-2) - 8\right)g(0)$ is ____ . [JEE (Adv.)-2024]
[Ans. : 51]

DIBY-234

If a function f satisfies $f(m+n) = f(m) + f(n)$ for all $m, n \in \mathbb{N}$ and $f(1) = 1$, then the largest natural number λ such that $\sum_{k=1}^{2022} f(\lambda + k) \leq (2022)^2$ is equal to:
[JEE Mains-2024]
[Ans. : 1010]

DIBY-235

For the differentiable function $f: \mathbb{R} - \{0\} \rightarrow \mathbb{R}$, let $3f(x) + 2f\left(\frac{1}{x}\right) = \frac{1}{x} - 10$, then $\left|f(3) + f'\left(\frac{1}{4}\right)\right|$ is equal to
[JEE Mains-2023]
(A) $\frac{33}{5}$ (B) 8 (C) $\frac{29}{5}$ (D) 13
[Ans. : (D)]

FUNCTIONAL EQUATIONS



DIBY-236

Let $5f(x) + 4f\left(\frac{1}{x}\right) = \frac{1}{x} + 3, x > 0$. Then $18 \int_1^2 f(x) dx$ is equal to **[JEE Mains-2023]**

- (A) $5 \log_e 2 + 3$ (B) $10 \log_e 2 + 6$ (C) $10 \log_e 2 - 6$ (D) $5 \log_e 2 - 3$
[Ans.: (C)]

DIBY-237

Let $f(x) + 2f\left(\frac{1}{x}\right) = x^2 + 5$ and $2g(x) - 3g\left(\frac{1}{2}\right) = x, x > 0$. If $\alpha = \int_1^2 f(x) dx$ and

$\beta = \int_1^2 g(x) dx$, then the value of $9\alpha + \beta$ is **[JEE Mains-2025]**

- (A) 1 (B) 0 (C) 10 (D) 11
[Ans.: (D)]

DIBY-238

For the function f , let $3f(x) + 2f\left(\frac{m}{19x}\right) = 5x, x \neq 0$ where $m = \sum_{i=1}^9 (i)^2$ then

$f(5) - f(2)$ is equal to **[JEE Mains-2026]**

- (A) 18 (B) 36 (C) 9 (D) -9
[Ans.: (A)]

FUNCTIONAL EQUATIONS



DIBY-239

Suppose that a function $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$ and $f(1) = 3$. If $\sum_{i=1}^n f(i) = 363$; then n is equal to ____.

[JEE Mains-2020]
[Ans.: 5]

DIBY-240

Let $\mathbb{R}$ denote the set of all real numbers. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function such that $f(x) > 0$ for all $x \in \mathbb{R}$, and $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Let the real numbers $a_1, a_2, \dots, a_{50}$ be in an arithmetic progression. If $f(a_{31}) = 64f(a_{25})$, and $\sum_{i=1}^{50} f(a_i) = 3(2^{25} + 1)$, then the value of $\sum_{i=6}^{30} f(a_i)$ is ____.

[JEE (Adv)-2025]
[Ans.: 96]

DIBY-241

Let $f: \mathbb{N} \rightarrow \mathbb{R}$ be a function such that $f(x + y) = 2f(x)f(y)$ for natural numbers x and y . If $f(1) = 2$, then the value of α for which $\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$ holds, is:

[JEE Mains-2022]

- (A) 2 (B) 3 (C) 4 (D) 6

[Ans.: (C)]

FUNCTIONAL EQUATIONS



DIBY-242

A function f is defined for all positive integers and satisfies $f(1) = 2002$ and $f(1) + f(2) + \dots + f(n) = n^2 f(n)$ for all $n > 1$. Find the value of $f(2004)$.

[Ans. :]

DIBY-243

Let f and g be functions satisfying $f(x + y) = f(x)f(y)$, $f(1) = 7$ and

$g(x + y) = g(x, y)$, $g(1) = 1$ for all $x, y \in N$. If $\sum_{x=1}^n \left(\frac{f(x)}{g(x)} \right) = 19607$, then n is equal to

[JEE Mains-2026]

(A) 7

(B) 6

(C) 4

(D) 5

[Ans. : (D)]

DIBY-244

Suppose f is a real function satisfying $f(x + f(x)) = 4f(x)$ and $f(1) = 4$. Find the value of $f(21)$.

[Ans. : 64]

DIBY-245

Let $f(x)$ be a function with two properties

- (i) For any two real number x and y , $f(x + y) = x + f(y)$ and
- (ii) $f(0) = 2$.

Find the value of $f(100)$.

[Ans. : 102]

*It is never about end result
It's always about Journey*



THANK
You

#futureellTians